



INSTITUTE OF INFORMATION TECHNOLOGY,
VIETNAM ACADEMY OF SCIENCE AND TECHNOLOGY

SOCIALIST REPUBLIC OF VIETNAM
Independence - Freedom - Happiness

INTERNSHIP CONFIRMATION

Intern's full name: Nguyen Cao Xuan Phuc

Email: nguyencxphuc@gmail.com

Tel: (+84) 034 9659560

Intern's position: Intern (Main Mechanical Designer & Hardware Developer)

Dates of Internship: 1st March - 15th September, 2026

Department: Department of Intelligent Systems and Advanced Computing

Address: No.18 Hoang Quoc Viet Street, Nghia Do Ward, Hanoi, Vietnam

Supervisor's name: Pham Van Khanh **Position:** Researcher, Member of the Institute's Science of Information Technology

Email: pvkhanh@ioit.ac.vn **ORCID:** 0000-0002-0227-4254

Project: Assistive Smart Glasses for the Visually Impaired (Sign Language-to-Speech Translation System)

In order to assess the students' internship and at the same time provide us with input for curricular improvement in the future, we would like you to give us your evaluation over the following criteria.

1. Mechanical Engineering & Design Mastery

1.1. CAD Modeling & Geometry Optimization

☒ Outstanding ☐ Good ☐ Average ☐ Needs improvement ☐ N/A

1.2. Applied Thermodynamics & Heat Dissipation

☒ Outstanding ☐ Good ☐ Average ☐ Needs improvement ☐ N/A

1.3. Structural Integrity & Material Selection



☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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1.4. Ergonomics & Human-Centered Design

☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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1.5. Design for Manufacturing & Prototyping (DFM)

☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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2. Mechatronics & Hardware Integration

2.1. Spatial Layout & Component Packaging

☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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2.2. Embedded Hardware Configuration

☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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2.3. System Troubleshooting & Empirical Debugging

☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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2.4. Design Iteration & Prototyping Speed

☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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3. Analytical Research & Problem-Solving

3.1. Technical Problem Formulation

☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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3.2. Empirical Data Analysis & Verification

☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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3.3. Engineering Adaptability & Innovation

☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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3.4. Technical Documentation & CAD Organization

☐ Outstanding	☒ Good	☐ Average	☐ Needs improvement	☐ N/A
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4. Professionalism & Soft Skills

4.1. Research Ownership & Self-Drive

☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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4.2. Cross-Disciplinary Technical Communication

☒ Outstanding	☐ Good	☐ Average	☐ Needs improvement	☐ N/A
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4.3. Empathy-Driven Engineering Mindset

☒ Outstanding ☐ Good ☐ Average ☐ Needs improvement ☐ N/A

4.4. Time Management & Milestone Execution

☒ Outstanding ☐ Good ☐ Average ☐ Needs improvement ☐ N/A

What specific technical contributions did Xuan Phuc make to the smart eyewear system, and how did his work impact the overall project?

Xuan Phuc served as the primary mechanical architect for the hardware chassis. He engineered the entire glasses frame from the ground up to accommodate a high-performance camera module, a Raspberry Pi processing unit, and audio output. His most critical breakthrough was designing a custom micro-thermal dissipation system. Running real-time computer vision AI models on embedded hardware generates significant heat; Xuan Phuc's passive cooling channels and thermal sink design lowered operational temperatures by over 15°C, enabling continuous operation without adding excess weight or sacrificing user safety.

How did Xuan Phuc demonstrate problem-solving capabilities when encountering hardware integration challenges?

During early prototype iterations, housing the Raspberry Pi board and camera sensor within a wearable form factor caused severe center-of-gravity imbalance and component overheating. Xuan Phuc systematically resolved this through iterative physical testing and CAD redesigns. He reconfigured the spatial orientation of the internal wiring and re-engineered the frame's weight distribution across the temple tips and nasal pads, achieving an optimal balance that ensured both structural stability and long-wear comfort.

How does Xuan Phuc's performance reflect his potential for top-tier undergraduate Computer Engineering studies in university?

Xuan Phuc possesses a rare combination of theoretical rigor and hands-on mechanical execution. He does not treat engineering as mere abstraction; he understands material properties, heat transfer principles, and manufacturing constraints intuitively. His human-centered mindset, combined with exceptional CAD proficiency and cross-disciplinary mechatronic execution, makes him an ideal candidate who will thrive in rigorous engineering environments at university.

Representative of Company
(Signed, full name and sealed)


TRƯỞNG PHÒNG
QUẢN LÝ TỔNG HỢP
Lại Quốc Anh

Hanoi, 15th September, 2026

Supervisor
(Signed, full name)


Pham Van Khanh